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Evaluating the Methodological Strengths and Limitations of 'Moderate Alcohol Intake Reduces Risk of Ischemic Stroke in Korea': A Paper Critique

Original Study by Soo Joo Lee *et al.*

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Study Aim and Design

Previous studies have consistently reported that heavy alcohol intake increases the risk of ischemic and haemorrhagic stroke. However, the effect of light-to-moderate drinking remains unclear and controversial, owing to heterogeneity in stroke subtypes, potential differential effects across sex and racial groups, differing beverage types, and amounts of alcohol consumed. To overcome these challenges in relation to elucidating association in Asian populations, Lee *et al.* performed a population-based case-control study aiming to examine a dose-relationship between alcohol intake and risk of ischemic stroke in Koreans (who had different alcoholic-beverage preferences than Western populations), considering potential effect-modifications by sex and ischemic stroke subtype.

The authors likely chose a population-based case-control approach due to its efficiency in examining multiple risk factors for ischemic stroke (IS), and the ability to explore dose-response relationships. While a prospective cohort study design would reduce biases and provide temporal evidence of causality, it would require significant cost in terms of time and resources. Moreover, a randomized controlled trial would be infeasible due to ethical considerations around assigning alcohol consumption. Other designs, such as ecological or cross-sectional studies, would not allow for individual-level inference or estimations of appropriate odds ratios (OR).

In contrast, nested case-controls and case-cohort designs could provide additional flexibility, efficiency, and stronger evidence about strength of association between alcohol consumption (exposure of interest) and IS (outcome), through temporality and estimations of other measures of association and impact. Despite alcohol consumption assessment via same structured questionnaire in both groups, case-interviews were prospectively conducted while control-interview data was retrospectively sourced. Nested designs would allow for exposure assessment in both groups before occurrence of IS, enabling temporality. However, practical constraints of conducting such studies in the Korean population may have made the chosen design a more feasible option for researchers at the time.

External and Internal Validity

An analytic observational design, the study offers good external validity within the Korean population, including scientific and statistical generalization which rest on biological and statistical representativeness, respectively. Internal validity, however, has been affected by measurement errors. In mitigation against random errors, Lee *et al.* included a large sample size, reducing variability and improving statistical power and precision of OR estimates. Violations of internal validity, however, stem from biases, which cannot be minimized by such.

Potential sources of non-response selection-bias arise from the exclusion of stroke patients who could not be interviewed due to severe conditions like altered consciousness or aplasia. This exclusion may lead to a sample that overrepresents milder stroke cases, potentially skewing the true relationship between alcohol-intake and IS. Additionally, Neyman's bias could occur due to underrepresentation of individuals with severe alcohol-related health issues, systematically distorting observed associations. For example, if excluded cases were more likely to engage in heavy drinking, the OR may overestimate the 'protective effect' of light-to-moderate drinking.

Efforts were made to address these biases by attempt to create a case-sample representative of the Korean stroke population (through use of CRCS-5 database which mirrored national demographics), and a control sample reflective of the population at risk (through use of KHANES which employed multistage-clustered probability sampling). Despite this, selection processes between both groups were not identical or independent of exposure status; cases were first-ever stroke patients, while controls were sourced from a health survey at a previous time. To better address these selection biases, the authors could have attempted to collect data on non-respondents through medical records to explore systematic differences, used a more comparable selection process for controls, and conducted sensitivity analyses to estimate potential impact of excluded cases.

Potential for information bias is another notable concern. The prospective nature of the case-interviews likely meant interviewers were not blinded to participant status, introducing risk of interviewer bias. Although the use of a standardized questionnaire across both groups minimizes differential bias, the reliance on retrospective self-report on alcohol consumption over the past 12-months exposes all participants to non-differential recall bias, which is further exacerbated by social-desirability bias. Proxy respondents for 29.9% of cases may also introduce differential bias, if proxies inaccurately reported consumption. Discrepancies in smoking definitions between cases and controls was also noted, which could confound results given the potential correlation between smoking and alcohol consumption. Finally, the temporal gap between case and control data-collection periods could have seen shifts in consumption patterns over time.

To address these limitations, the authors could have supplemented self-reports with objective biomarkers of alcohol consumption, such as blood or hair ethanol metabolites. Validation studies comparing self and proxy reports against biomarkers would help quantify reporting inaccuracies, while using shorter recall periods or diary methods could improve reporting accuracy. These measures would strengthen the reliability of findings while reducing the impact of information bias.

Confounding

Internal validity can also be threatened by confounding. The crude OR (95% CIs) of 1.35 (1.06-1.71) indicates the odds of IS are ~35% higher among those who consume 5-6 drinks on average compared to non-drinkers. As the CI does not include 1, this suggests a statistically significant association. The adjusted OR of 1.23 (0.93-1.63) provides a more realistic estimate, reducing the apparent effect of alcohol by accounting for confounding risk factors. Though not statistically significant, the wider interval is likely due to inclusion of multiple variables, reducing statistical power. Crucially, adjusting for the higher frequency of confounding variables in cases compared to controls would attenuate the association between heavy drinking and stroke risk.

The 8.89% difference in ORs suggests a small part of the crude association between exposure and outcome was influenced by confounding factors, which were more common in cases than controls. Potential sources of these factors identified in the study include demographic considerations, traditional stroke factors, and beverage-type. The authors addressed these by matching cases and controls on sex, age, and education level to control for confounding at the design-phase, and adjusting for hypertension, hyperlipidaemia, diabetes, coronary heart disease, bmi, and current-smoking status via conditional logistic regression at the analysis-phase.

Although the paper does not provide specification around the identification of these 'confounders', they were likely chosen based on existing literature and biological plausibility. However, it is unclear whether crude and adjusted ORs were compared for all categories of alcohol consumption, ideally demonstrating a difference of 15% or more between estimates to be confident of confounding. Importantly, the omission of atrial fibrillation due to lack of data in the control database, and inconsideration of other potential factors such as diet and physical activity, could introduce residual confounding, given they may influence both exposure and outcome. While matching could have increased internal validity and statistical-power of the study, doing so for variables that are not true confounders would reduce statistical-efficiency. As such, the authors could have benefitted collecting more comprehensive data on socio-economic and lifestyle factors to further reduce residual confounding and improve the study's validity.

Post-hoc analyses of drinking patterns, and stratification by sex and stroke subtypes provided additional insights but did not fully eliminate the potential for confounding. Notably, beverage-type was acknowledged as a potential confounder, but the lack of comparable data for controls limited analysis. To address these limitations, the study could have employed sensitivity analyses, biomarkers, or additional covariates to capture unmeasured confounders like physical activity or socio-economic status.

Effect Modification

These post-hoc analyses also explored the possibility for effect-modification. Stratification by sex revealed the protective effect of light-to-moderate drinking persisted up to 3-4 drinks per day in men, but only up to 1-2 day in women. However, this interaction between sex and alcohol intake was not statistically significant (p=0.86). Similarly, no interaction was found when stratifying by IS-subtype, as indicated by a p-value greater than 0.05 (p=0.50). These results suggest consistency in the association across sexes and stroke subtypes, providing little evidence of effect-modification in these contexts.

Dichotomization of participants into regular and irregular drinkers also enabled additional insights. The authors found the protective effect of moderate drinking (3-4 drinks/day) was observed only in regular drinkers, suggesting potential modification by drinking frequency. This analysis, however, was exploratory, and not robust enough to conclude effect-modification. The study could have further explored effect-modification through inclusion of interaction terms in regression models, designed to test for such modifications by variables like drinking patterns, sex, or age. Synergism and antagonism could be investigated in these models to elucidate identified effects. By integrating these methods, the study's design could have more rigorously identified and reported any true effect-modifiers – factors which cannot and should not be ignored.

Conclusion

Overall, the authors have provided a partially satisfactory answer to their research question, elucidating the dose-response relationship between alcohol consumption and ischaemic stroke in Koreans while exploring sex and stroke subtypes as potential effect-modifiers. They demonstrated significant associations for light-to-moderate drinking and ischaemic stroke, but found no statistically significant effect-modification by sex or stroke subtype. Notably, reliance on retrospective self-reports, potential bias and confounding, and the lack of temporality in the study design limit the strength of findings. While the study highlights plausible biological mechanisms and considers relevant confounders, unmeasured factors like socio-economic status and physical activity, and the dis-inclusion of biomarkers, leave residual uncertainties.

Original Study

Lee SJ, Cho YJ, Kim JG, Ko Y, Hong KS, Park JM, et al. Moderate alcohol intake reduces risk of ischemic stroke in Korea. Neurology. 2015 Dec;85(22):1950–6.